

Structured Prompt Template

Structured Prompt Template is a structured script that is sent to an AI model like OpenAI GPT every time I ask a question.

Main Components in Structured Prompt:

Here are 4 components to ensure AI doesn't give random answers and stays focused:

1. System prompt

The system prompt is an absolute instruction that sits at the top of the system. Here's the code that will dictate the AI's character:

- **Persona:** You are ShopAssist, a friendly and helpful virtual shopping assistant.
- **Limitation:** You may only answer questions about products listed in the catalog. If a user asks about politics or cooking recipes, answer politely.
- **Format:** Use bold Markdown formatting for product names and always include the price.

2. Context Injection

Data from products.json processed in the Repository is entered, the code will convert a set of product data into neat text and insert it into the prompt.

This step is crucial because the AI model doesn't inherently know what items are being sold. By inputting this data, the AI will be able to retrieve the correct product data..

3. Chat History

The system will read past conversations and take user messages and previous AI replies and incorporate them into the prompts. This will make the conversation feel fluid and natural, as the AI remembers the context of the conversation from several steps back.

4. Visual Context

Visual Context is a feature in my app.py file. Because I'm using a vision-enabled model like gpt-4o-mini, the prompt isn't just text. If the user uploads an image, the code converts it to Base64 and inserts it into the user's question.

Structured Prompt Template is essential for assembling prompts with detailed structure:

- **Prevent hallucinations:** The AI won't recommend "MacBook Pro" if only "Huawei Laptop" and "ASUS Laptop" are available. It will rely on the data provided in the Context section.
- **Consistency of tone of language:** AI will always sound like a shopping assistant because its character has been programmed in the System Prompt.
- **Token Efficiency:** By organizing what is necessary and what is not, OpenAI API costs will be saved.